increased demand for passenger and cargo aircraft, Modi said in Vadodara that India would need more than 2,000 aircraft in the next 15 years.

According to an Allied Market Research report, the value of the global military transport aircraft is set to touch $45 billion by 2030. The Tata-Airbus facility is due to complete its IAF commitment by 2031 and is free to export aircraft after that.

Now that the government has opened up aircraft manufacturing to private players, policymakers will possibly not engage HAL for all their needs, thus putting all their eggs in one basket, and distribute the workload amongst private players. Just like Tata has been chosen for transport jets, another firm might be shortlisted to make fighter aircraft. On similar lines, manufacture of helicopters and UAVs (unmanned aerial vehicles) could be entrusted to separate companies.

From the Russian origin Sukhoi (manufactured under licence) to the indigenously designed and developed LCA Tejas, HAL has been making fighter jets for IAF. Besides fighter aircraft, HAL produces multiple types of helicopters for the armed forces. Its hands are full too, with close to 1,000 helicopters to be delivered in the coming years. Recently, the Light Combat Helicopter, Prachand, also made by HAL, was inducted into the IAF. The order to manufacture HTT-40 trainer jets was also given to HAL recently. In addition, the IAF is committed to acquire over 300 LCA Tejas fighter jets—a 4.5 generation fighter. While 40 LCA Mk1 are being delivered, 83 upgraded Mk1A versions have been ordered by the IAF. LCA Mk2, a medium weight fighter, will likely have its first flight by late 2023 or early 2024. All of them are manufactured by HAL.

It's the workload that military observers believe is behind delays in delivery of aircraft by HAL. They think it is high time HAL shared its burden. "Being the only aircraft manufacturer, HAL's plate is full. And it has led to repeated delays in delivery. From LCA Tejas to helicopters, HAL's record of timely delivery is not up to the mark. It should offload or share tasks by joining hands with private players," says a key defence official. Even some IAF officials have openly criticised HAL for missing deadlines. However, it has to be admitted that HAL is hampered by long-winded bureaucratic (MoD) processes in the award of contracts.

HAL, or Hindustan Aeronautics Ltd, as it was known then, started manufacturing aircraft in 1942, with licensed production of the American fighter aircraft Curtiss P-36 Hawk and dive bomber Vultee A-31 Vengeance for the Royal Indian Air Force. Over the years, it has produced (under licence) de Havilland Vampires, MiG-21 variants, Folland Gnats, MiG-27 M variants, SEPECAT Jaguars, BAE Hawks and Sukhoi Su-30MKIs.

At times, private Indian firms have appealed to the government to show confidence in their capabilities, so that they, too, can contribute towards attaining self-reliance in military hardware. Air Vice Marshal Anil Chopra (retd), director general, Centre for Air Power Studies, takes note of the momentous first that the Tata-Airbus deal represents. "First, transport jets will be made in India and, second,

HAL HAS ITS HANDS FULL WITH PROJECTS. IT IS TIME THAT IT SHARED ITS BURDEN WITH PRIVATE FIRMS

C-295

WAGONS IN THE SKY

THE TATA-AIRBUS JOINT VENTURE TO MANUFACTURE THE C-295 TRANSPORT AIRCRAFT FOR THE IAF WILL BE A FIRST FOR AN INDIAN PRIVATE FIRM, AND CAN WELL BE A SIGN OF THINGS TO COME. A FEW FEATURES OF THE ₹21,935 CRORE PROJECT:



- C-295MW is a transport aircraft of 5-10 tonne capacity with a maximum speed of 480 kmph. It will replace the ageing Avro aircraft of IAF. It has a rear ramp door for quick reaction and para-dropping of troops and cargo. Short take-off/ landing from semi-prepared surfaces is another of its features.
- The aircraft will strengthen the logistical capabilities of the IAF
- The first 16 flyaway aircraft are scheduled to arrive between September 2023 and August 2025. The first Made in India aircraft is expected in September 2026
- MoD officials claim that 96 per cent of the total man-hour work per aircraft that Airbus employs at its manufacturing facility in Spain will be undertaken in India by the Tata Advanced Systems Ltd. Manufacturing of over 13,400 detail parts, 4,600